# ASIC Physical Design Flow Workshop

## Complete Stage-by-Stage Guide with Specifications, Real-Time Diagrams & Tapout Introduction

| # | Stage | Tool | Output |
|---|---|---|---|
| 1 | RTL Design | VCS / Verdi / ModelSim | Verified Verilog RTL |
| 2 | Synthesis | Design Compiler / Genus | Gate-Level Netlist |
| 3 | Floorplan | ICC2 / Innovus | Floorplan DEF |
| 4 | Placement | ICC2 / Innovus | Placed Netlist |
| 5 | CTS | ICC2 / Innovus | Balanced Clock Tree |
| 6 | Routing | ICC2 / Innovus | Routed Layout |
| 7 | STA | PrimeTime / Tempus | Timing Sign-off |
| 8 | DRC / LVS | Calibre / PVS | Clean GDS |
| 9 | Tapout | Calibre + Foundry | GDS II to Fab |

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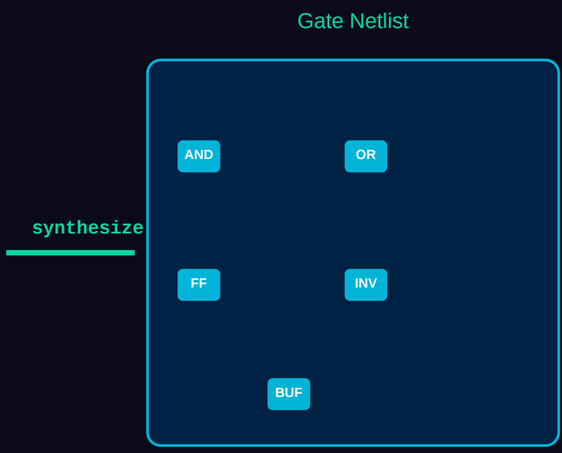
JNTU Anantapur · SiValley Technologies Internship · 28nm / 32nm

# STAGE 1 RTL Design & Verification

RTL (Verilog)

```
module top(
  input clk,
  output [7:0] q
);
always @(posedge clk)
  q <= d + 1;
endmodule
```




## Overview

RTL (Register Transfer Level) is the entry point of ASIC design. Engineers describe hardware behavior using HDLs such as Verilog or VHDL. The RTL code models data flow between registers and the logic operations performed. Functional verification using simulation (VCS, ModelSim) and formal methods ensures correctness before proceeding. A clean RTL with proper coding guidelines (no latches, synchronous resets, CDC handled) is critical for synthesis quality. In a real project at 28nm (e.g., ORCA TOP), RTL is received from design architects and the physical design team performs RTL reviews for synthesis readiness — checking clock domains, reset structure, and timing-critical paths.

## Specifications & Parameters

| | |
|---|---|
| **Technology Node** | 28nm / 32nm (TSMC / Intel) |
| **HDL Language** | Verilog / SystemVerilog / VHDL |
| **Simulation Tool** | Synopsys VCS, Cadence Xcelium, ModelSim |
| **Formal Verify** | Cadence JasperGold, Synopsys VC Formal |
| **Clock Domains** | Single / Multi-CDD with CDC analysis |
| **Reset Strategy** | Synchronous Active-Low preferred |
| **Lint Check** | Spyglass / Ascent Lint |
| **Coverage Target** | Line: 100% \| Branch: 95%+ \| Toggle: 90%+ |
| **CDC Analysis** | 0 violations — handshake / sync flops required |
| **File Format** | .v / .sv / .vhd / .f (filelist) |

## Real-Time Tool Commands (ICC2 / Industry)

```
# VCS Simulation
vcs -full64 -sverilog top.v tb.v -o simv
./simv +TESTNAME=smoke_test
# Spyglass Lint
spyglass -top top_module -lib stdcell.db -goal lint/lint_rtl
# CDC Analysis
spyglass -goal cdc/cdc_verify -top top_module
```

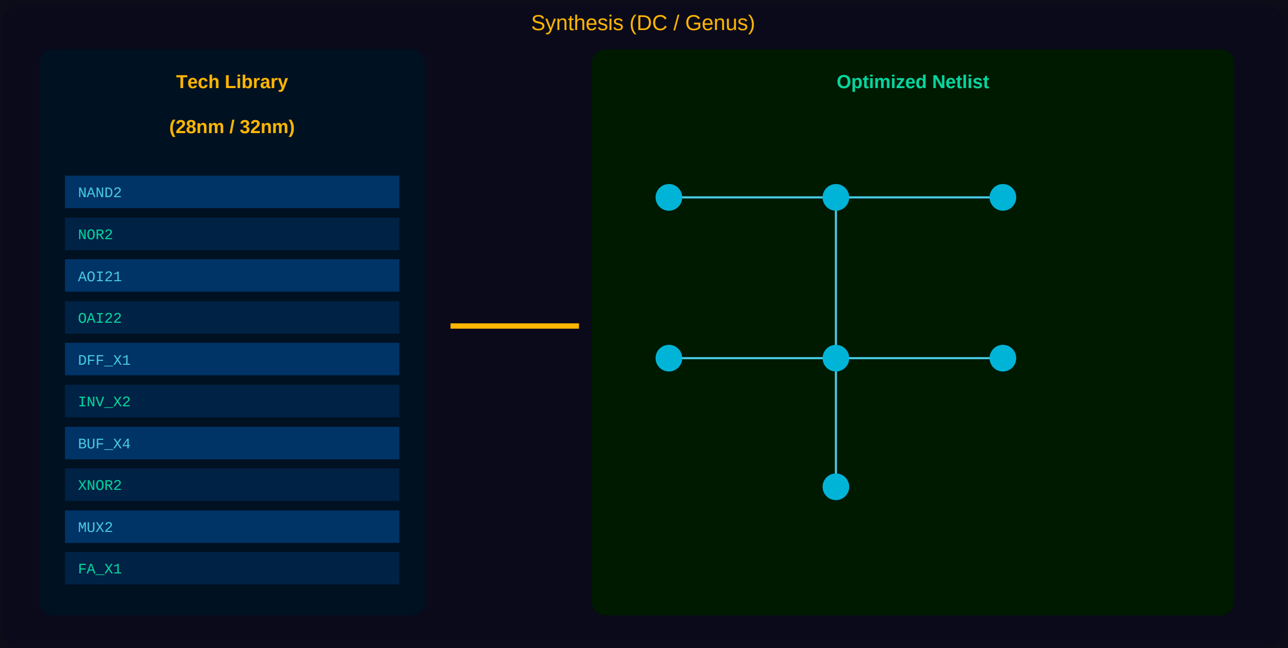
## Real-Time Engineer Notes

- *Always check for unintended latches — use 'always_ff' and 'always_comb' in SystemVerilog.*
- *At 28nm, clock gating cells must be instantiated explicitly — avoid RTL-level gating.*
- *Review reset fanout — excessive reset trees cause CTS congestion later.*
- *CDC violations found at RTL save 3–4 weeks of post-layout debugging.*
- *ORCA TOP: 8 clock domains, synchronous reset, 120K standard cells post-synthesis.*

## Tapout Relevance

RTL quality directly determines tapout success. Un-synthesizable RTL, CDC issues, or reset glitches discovered post-placement can delay tapout by months. A clean RTL sign-off at this stage — lint-clean, CDC-clean, simulation coverage >95% — is the first gate in the tapout checklist.

# STAGE 2 Logic Synthesis




## Overview

Logic Synthesis converts RTL into a technology-mapped gate-level netlist using a standard cell library. The synthesis tool (Design Compiler / Genus) maps Boolean logic to cells from the PDK library and optimizes for timing (meeting setup/hold constraints), area (cell count), and power (switching activity). Constraints are provided via SDC (Synopsys Design Constraints) — including clock definitions, input/output delays, false paths, and multicycle paths. The output is a gate-level netlist (.v) and reports for timing, area, and power. At 28nm in ICC2 projects like ORCA TOP, synthesis uses the TSMC 28HPC library with multiple Vt options (SVt, LVt, HVt).

## Specifications & Parameters

| | |
|---|---|
| **Tool** | Synopsys Design Compiler (DC) / Cadence Genus |
| **Technology Library** | TSMC 28nm HPC / SMIC 32nm |
| **Cell Library Format** | .db (Synopsys) / .lib (Liberty) |
| **Constraint Format** | SDC (Synopsys Design Constraints) |
| **Target Frequency** | 500 MHz – 1 GHz (design dependent) |
| **WNS Target** | 0 ps (no negative slack) |
| **TNS Target** | 0 ps total negative slack |
| **Area Budget** | Design specific — mm² or gate count |
| **Power Estimation** | Dynamic + Leakage — SAIF/FSDB toggle rates |
| **Output** | Gate netlist .v, SDF, SDC, area/timing/power reports |

## Real-Time Tool Commands (ICC2 / Industry)

```
# DC Shell — Synthesis
set_db hdl_search_path ./rtl
read_hdl -sv top.sv
elaborate top
read_sdc constraints.sdc
syn_generic; syn_map; syn_opt
write_hdl > netlist.v
```

```
report_timing -path full > timing_report.txt
report_area > area_report.txt
report_power > power_report.txt
```

### Real-Time Engineer Notes

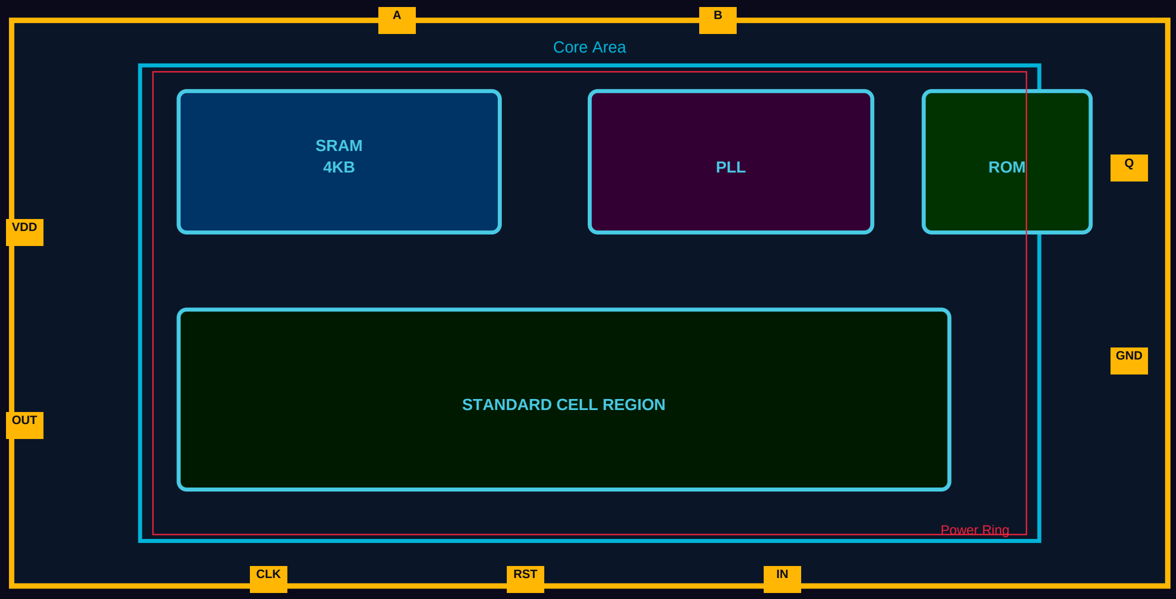
- *Always compile with '-scan' option if the design has DFT scan chains.*
- *Use 'set_max_fanout' and 'set_max_transition' to control signal integrity early.*
- *Check setup WNS — if > -200ps, resolve before physical design; else fix in placement.*
- *At 28nm: HVt cells save power, LVt cells meet timing — mix Vt for optimal PPA.*
- *ORCA TOP had 3.2% negative slack post-synthesis — resolved during ICC2 post-route opt.*

### Tapout Relevance

A clean synthesis netlist with 0 WNS is the foundation for timing closure. Incorrect SDC constraints propagate into physical design and cause tapout-blocking timing issues. The synthesis sign-off report (timing, area, DFT coverage) is a mandatory tapout pre-requisite checklist item.

# STAGE 3 Floorplanning

Floorplan — Die / Core / Macros / IO




## Overview

Floorplanning establishes the physical framework of the chip. It involves defining the die size, core area, IO pad placement, macro placement (SRAMs, PLLs, analog blocks), and power delivery network (PDN). The core utilization ratio (typically 60–80%) balances congestion and area. Power rings and power stripes are drawn to deliver VDD/VSS across the die. Macro placement is critical — macros should be placed at die edges or corners to minimize routing blockages and halos must be set to prevent cell placement congestion. In ICC2, the floorplan is defined using TCL commands or GUI, and DEF/LEF files capture the physical layout.

## Specifications & Parameters

| | |
|---|---|
| **Tool** | Synopsys ICC2 / Cadence Innovus |
| **Die Size** | Design-specific (e.g., 2.5mm x 2.5mm at 28nm) |
| **Core Utilization** | 60–75% (standard cells) |
| **IO Ring** | Pad-limited vs Core-limited selection |
| **Macro Types** | SRAM, ROM, PLL, Analog IP, PHY |
| **Macro Halo** | 5–10 µm on each side |
| **Power Grid** | VDD/VSS rings + horizontal/vertical stripes |
| **Metal Layers for PG** | M7–M9 (top metals, low resistance) |
| **Row Orientation** | Alternating N/S for well sharing |
| **Output Files** | DEF, floorplan.tcl, PG netlist |

## Real-Time Tool Commands (ICC2 / Industry)

```
# ICC2 Floorplan Commands
initialize_floorplan -core_utilization 0.70 -aspect_ratio 1.0
create_die_area -boundary {{0 0} {2500 2500}}
place_io_pads -file io_pad_constraints.tcl
place_macro -macro_name SRAM_4KB -location {150 1500} -orientation R0
create_pg_ring -net VDD -layer M8 -width 10 -spacing 2
create_pg_stripe -net VDD -layer M7 -width 4 -pitch 50
```

```
connect_pg_net -net VDD -domain PD_CORE
check_floorplan > floorplan_check.rpt
```

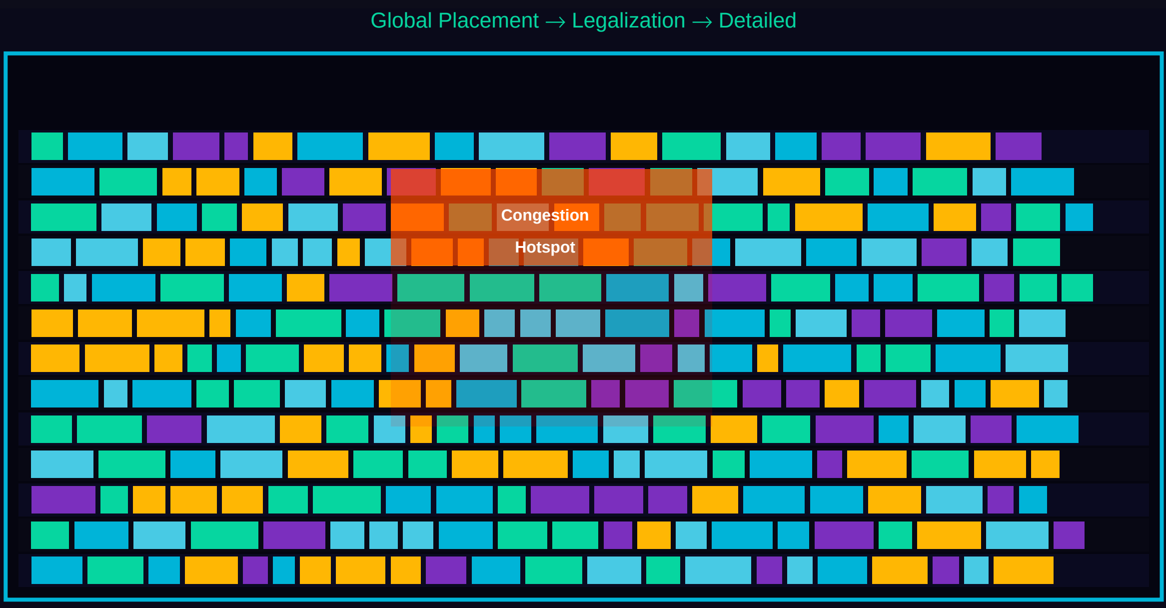
### Real-Time Engineer Notes

- *Always set macro halos before global placement — missing halos cause severe placement congestion.*
- *Verify PG connectivity — open nets in power grid cause IR drop failures and tapout rejection.*
- *Keep analog blocks (PLL, ADC) away from digital switching regions to avoid noise coupling.*
- *At 28nm, well-tap cells must be placed within 25µm to meet latch-up rules.*
- *ORCA TOP: PLL placed top-right, SRAM at bottom — reduced routing congestion by 15%.*

### Tapout Relevance

Floorplan quality determines routing congestion and IR drop — both are tapout sign-off criteria. A bad macro placement discovered after routing can require a full re-floorplan, adding weeks. Power grid must pass IR drop analysis (< 5% supply drop) before tapout approval from the foundry.

# STAGE 4 Placement




## Overview

Placement assigns physical locations to all standard cells within the core area. It occurs in three phases: global placement (coarse, minimizing wire length), legalization (snapping cells to rows, resolving overlaps), and detailed placement (fine-tuning for timing/congestion). The tool optimizes for timing (setup/hold), congestion (routing demand vs capacity), and power. Post-placement, timing analysis (in-context with estimated parasitics) identifies critical paths. In ICC2, placement is done with 'place_opt' which simultaneously optimizes timing and congestion. At 28nm, placement with scan chain ordering and clock gating awareness is critical.

## Specifications & Parameters

| | |
|---|---|
| **Tool** | ICC2 place_opt / Innovus place_design |
| **Placement Mode** | Timing-Driven + Congestion-Driven |
| **Target Utilization** | 65–75% (standard cell region) |
| **WNS after Placement** | < -100 ps (acceptable for pre-CTS) |
| **Congestion Metric** | GRC overflow < 1% on all layers |
| **Scan Chain Reorder** | Enabled for DFT |
| **Clock Gating** | Integrated clock gating cells placed |
| **Estimated RC** | Virtual route / steiner tree parasitics |
| **Output** | Placed DEF, timing reports, congestion maps |
| **Timing Mode** | Pre-CTS (ideal clocks) |

## Real-Time Tool Commands (ICC2 / Industry)

```
# ICC2 Placement
place_opt -effort high
legalize_placement
refine_placement -effort high
report_timing -max_paths 100 > pre_cts_timing.rpt
report_congestion -layers {M1 M2 M3} > congestion.rpt
check_placement -verbose > placement_check.rpt
```

```
write_def -output placed.def
```

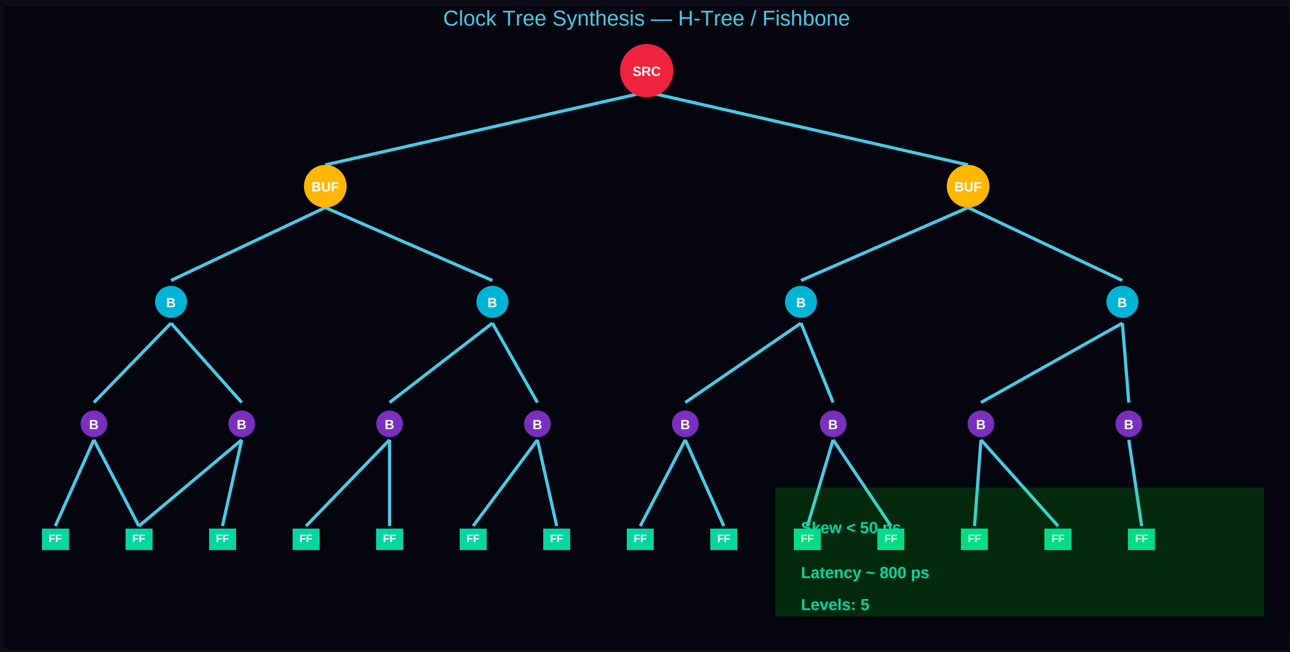
### Real-Time Engineer Notes

- *Monitor GRC (Global Route Congestion) during placement — hotspots > 5% overflow need macro adjustment.*
- *High fanout nets (reset, scan_en) need buffer insertion before placement to avoid timing failures.*
- *Use 'set_dont_touch' on critical IPs and macros to prevent unwanted optimization moves.*
- *Check hold violations even pre-CTS — they worsen after clock tree insertion.*
- *Raven Wrapper (32nm): placement utilization was 72% — congestion resolved by splitting scan chains.*

### Tapout Relevance

Post-placement timing and congestion reports are reviewed in design reviews before CTS. Excessive congestion at this stage predicts routing DRC violations — a key tapout blocker. Placement sign-off includes: WNS < 0 (acceptable pre-CTS), congestion maps reviewed, scan chain verified.

# STAGE 5 Clock Tree Synthesis (CTS)




## Overview

CTS builds the physical clock distribution network from the clock source to all sequential elements (flip-flops, latches). The goal is to minimize clock skew (arrival time difference between FFs) and control clock latency (source to sink delay). The clock tree uses buffers and inverters (often special CTS cells with less variation) arranged in H-tree or fishbone topologies. In ICC2, 'clock_opt' performs CTS and post-CTS optimization together. After CTS, the design switches from ideal clock mode to propagated clock mode for accurate timing analysis. At 28nm, multi-corner multi-mode (MCMM) CTS ensures skew meets across all PVT corners.

## Specifications & Parameters

| | |
|---|---|
| **Tool** | ICC2 clock_opt / Innovus ccopt_design |
| **Target Skew** | < 50 ps (local) / < 100 ps (global) |
| **Clock Latency** | 500 ps – 2 ns (source to farthest FF) |
| **Tree Topology** | H-Tree / Modified H-Tree / Fishbone |
| **CTS Cells** | CLKBUF_X4, CLKBUF_X8, CLKINV_X4 |
| **Max Transition** | 150 ps on clock nets at 28nm |
| **Max Capacitance** | 50 fF per clock buffer output |
| **Levels in Tree** | 4–6 (depending on sink count) |
| **Clock Domains** | Handled per domain with virtual clocks |
| **Post-CTS Mode** | Propagated clocks for STA |

## Real-Time Tool Commands (ICC2 / Industry)

```
# ICC2 CTS
set_clock_tree_options -target_skew 50 -target_latency 1000
clock_opt -only_cts
clock_opt -only_psyn
report_clock_qor > cts_qor.rpt
report_clock_tree -summary > clock_tree.rpt
check_clock_tree -clock CLK > cts_check.rpt
```

```
report_timing -path_type full_clock > post_cts_timing.rpt
```

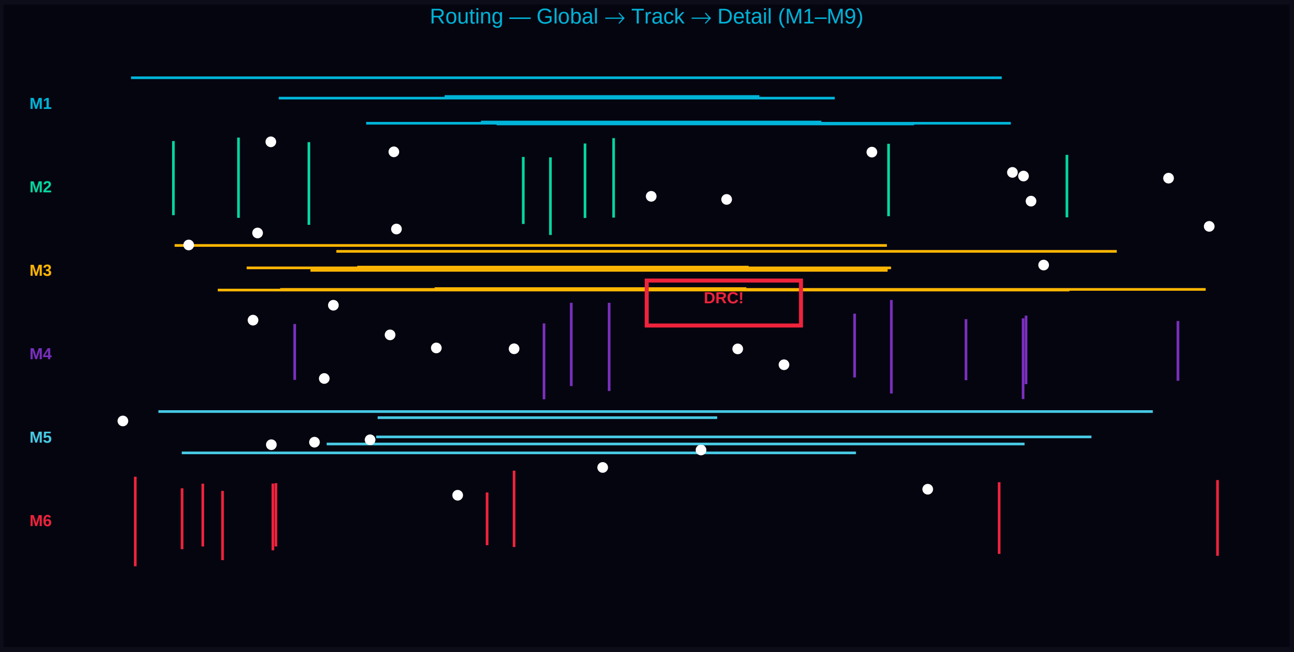
## Real-Time Engineer Notes

- *Always fix max_transition violations on clock nets — they cause functional failures across PVT corners.*
- *Check clock shielding requirements at 28nm — clock nets need shielding to reduce crosstalk noise.*
- *Hold violations dramatically increase post-CTS — use 'hold_fixing' during clock_opt.*
- *CTS cells should use HVt for power saving — confirm with your sign-off corner requirements.*
- *ORCA TOP: CTS achieved 38ps skew across 4500 flip-flops using 5-level H-tree.*

## Tapout Relevance

CTS quality is directly tied to setup and hold closure — timing sign-off is impossible without a balanced clock tree. Skew > 100ps causes hold violations that require buffer insertion in routing, increasing congestion. CTS QOR report (skew, latency, DRVs) is a mandatory tapout checklist item.

# STAGE 6 Routing




## Overview

Routing connects all signal and clock nets using physical metal wires on the available routing layers. It proceeds in phases: global routing (assigns nets to routing regions), track assignment (assigns nets to specific tracks), and detailed routing (generates actual metal shapes with DRC compliance). At 28nm, designs use 9–10 metal layers (M1–M9/AP). Lower layers (M1–M3) are for local cell connections, middle layers (M4–M6) for signal routing, and upper layers (M7–M9) for power. Routing must satisfy DRC rules (min spacing, min width, min area, antenna rules) and timing constraints simultaneously.

## Specifications & Parameters

| | |
|---|---|
| **Tool** | ICC2 route_opt / Innovus routeDesign |
| **Metal Layers** | M1–M9 + Alucap (AP) at 28nm |
| **Routing Direction** | H/V alternating per layer (M1:V, M2:H, M3:V...) |
| **Min Metal Width** | M1: 0.065µm \| M2: 0.07µm \| M7: 0.4µm |
| **Min Metal Spacing** | M1: 0.065µm \| DPT considerations at 28nm |
| **Via Types** | Single / Array vias — via optimization enabled |
| **DRC Target** | 0 violations post-route |
| **Antenna Rule** | Ratio < 400 (M1), < 200 (higher layers) |
| **Signal Integrity** | Crosstalk delta delay < 5% of slack |
| **Output** | Routed DEF, GDS (via stream-out), DRC reports |

## Real-Time Tool Commands (ICC2 / Industry)

```
# ICC2 Routing
route_auto -max_detail_route_iterations 10
route_opt -effort high
check_drc -check_via > post_route_drc.rpt
report_timing -max_paths 500 > post_route_timing.rpt
fix_antenna_violations -method diode_insertion
verify_route -antenna > antenna_check.rpt
```

```
write_gds -output final_routed.gds
```

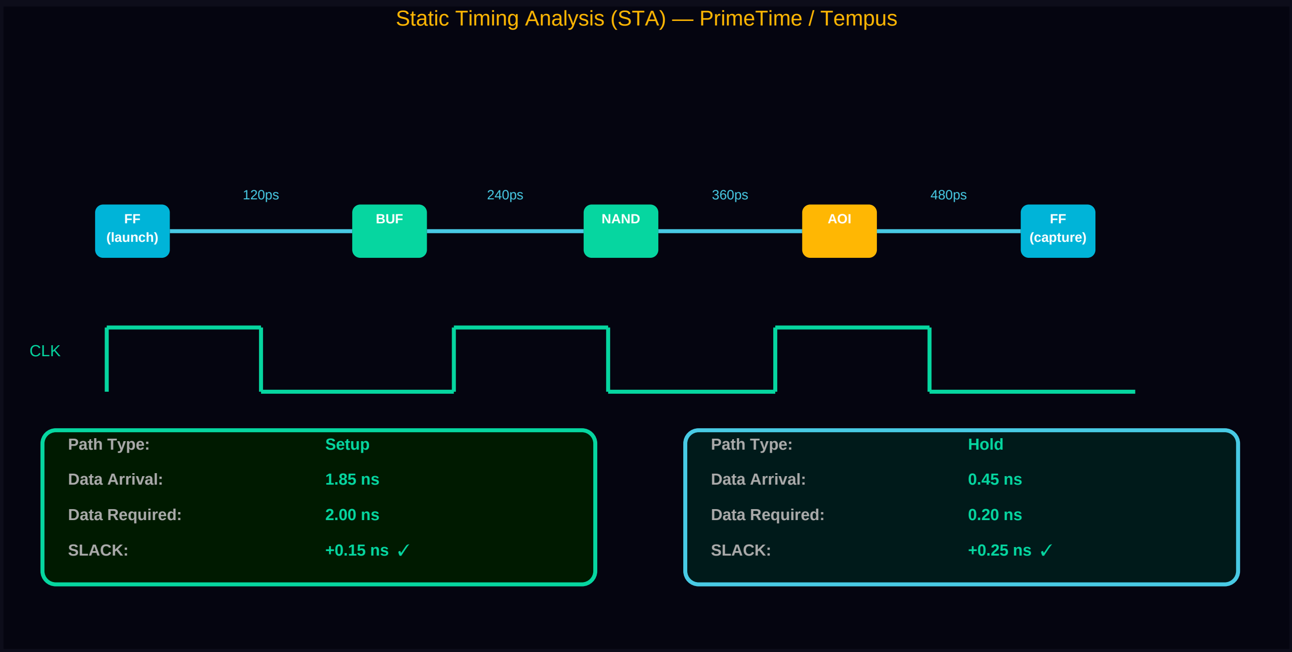
### Real-Time Engineer Notes

- *Floating wire DRCs (ORCA TOP experience) — caused by incomplete routes; run 'check_routes' to find open nets.*
- *At 28nm, double-patterning (DPT) constraints on M2/M4 require coloring — run DPT sign-off after routing.*
- *Antenna violations on long clock nets — insert diode cells or jumper via higher layers.*
- *Post-route SI analysis: use Synopsys StarRC for parasitics extraction before final STA.*
- *Raven Wrapper: 3 shorts found post-route in M3 — fixed by rerouting 2 critical signal buses.*

### Tapout Relevance

Routing is the last major physical design step before verification. The routed GDS must achieve 0 DRC violations — foundries reject GDS with any DRC errors. Routing sign-off covers: DRC clean, antenna violations fixed, SI analysis complete, LVS net-list matching GDS.

# STAGE 7 Static Timing Analysis (STA)




## Overview

STA verifies that all timing constraints are met in the physical design without requiring functional simulation. It analyzes every combinational path in the design for setup (max delay) and hold (min delay) violations. Using extracted parasitics (from StarRC / QRC) and Liberty models at multiple PVT corners, STA provides accurate timing closure data. Multi-mode multi-corner (MCMM) STA covers functional mode, scan shift mode, and low-power mode at fast/slow/typical corners. PrimeTime (Synopsys) or Tempus (Cadence) are industry standard STA tools. Sign-off STA must show 0 WNS and 0 TNS across all modes and corners.

## Specifications & Parameters

| | |
|---|---|
| **Tool** | Synopsys PrimeTime / Cadence Tempus |
| **Parasitics** | StarRC (Synopsys) / QRC (Cadence) — SPEF format |
| **Analysis Corners** | SS/0.81V/125°C \| FF/0.99V/-40°C \| TT/0.9V/25°C |
| **Setup Target WNS** | 0 ps (all modes, all corners) |
| **Hold Target WNS** | 0 ps (all modes, all corners) |
| **TNS Target** | 0 ps |
| **Analysis Modes** | Func / Scan_Shift / Low_Power / Test |
| **SI Noise Mode** | AOCV / SOCV / POCVM (advanced OCV) |
| **Clock Uncertainty** | Setup: 50 ps \| Hold: 0 ps |
| **Output** | Timing reports, SPEF, ECO scripts |

## Real-Time Tool Commands (ICC2 / Industry)

```
# PrimeTime STA
read_db tsmc28nm_ss.db
read_verilog routed_netlist.v
read_sdc constraints.sdc
read_parasitics -format spef routed.spef
set_operating_conditions -max ss_0p81v_125c
update_timing -full
```

```
report_timing -max_paths 100 -slack_lesser_than 0 > wns_report.txt
report_global_timing > global_timing_summary.txt
# ECO for timing fix
pt_shell> fix_eco_timing -setup -type size_cell
```

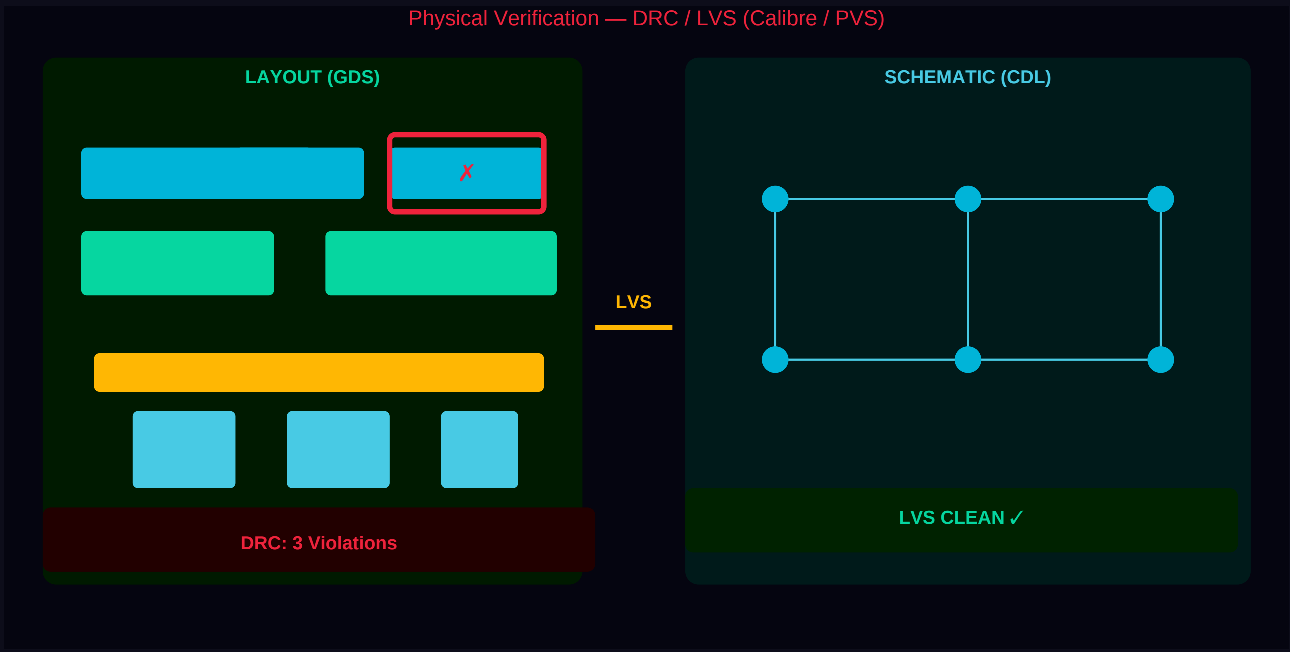
### Real-Time Engineer Notes

- *AOCV (Advanced OCV) derating adds realistic variation margin — always use for sign-off, not LOCV.*
- *Check hold violations at the fastest corner (FF/-40C/0.99V) — hold ECOs must not create setup violations.*
- *Long interconnect paths at 28nm: use SPEF with coupling caps — ignoring them underestimates crosstalk delay.*
- *False paths and multicycle paths in SDC must be reviewed — incorrect SDC causes fake timing violations.*
- *ORCA TOP STA: 12 setup ECOs applied (cell sizing + buffer insertion) to achieve timing closure at SS corner.*

### Tapout Relevance

STA sign-off is the most critical tapout gate. The foundry and IP vendors require STA sign-off reports demonstrating 0 WNS / 0 TNS across all corners and modes. Any negative slack at sign-off means the chip will not function at its rated frequency and cannot be submitted for fabrication.

# STAGE 8 Physical Verification — DRC & LVS




## Overview

Physical verification ensures the layout (GDS) is manufacturable (DRC) and electrically equivalent to the schematic/netlist (LVS). DRC (Design Rule Check) verifies that all geometric constraints from the foundry Process Design Kit (PDK) are met — spacing, width, overlap, density, antenna ratios. LVS (Layout vs Schematic) compares the extracted netlist from GDS against the reference gate-level netlist to confirm they match. Calibre (Mentor/Siemens) is the industry-standard tool. Additional checks include ERC (Electrical Rule Check), density fill verification, and DPT (Double Patterning) coloring verification at 28nm.

## Specifications & Parameters

| | |
|---|---|
| **DRC Tool** | Siemens Calibre nmDRC / Synopsys ICV |
| **LVS Tool** | Siemens Calibre nmLVS / Synopsys PVS |
| **Rule Deck** | Foundry-provided PDK rule file (e.g., TSMC28_DRC.cal) |
| **DRC Target** | 0 violations — clean GDS submission |
| **LVS Target** | 0 errors — nets, devices, pins must match |
| **ERC Checks** | Floating gates, undriven nets, power shorts |
| **Density Rules** | Metal fill for uniform density (TSMC: 30–80%) |
| **Antenna Check** | Antenna ratio < 400 (process dependent) |
| **DPT Coloring** | 28nm: M2/M4 double-patterning conflict-free |
| **Output** | DRC summary report, LVS comparison report, clean GDS |

## Real-Time Tool Commands (ICC2 / Industry)

```
# Calibre DRC
calibre -drc -hier -turbo 8 -hyper tsmc28_drc.cal -spice netlist.sp layout.gds
# Calibre LVS
calibre -lvs -hier -turbo tsmc28_lvs.cal -spice schematic.cdl layout.gds
# Calibre DRV (Density)
calibre -drv density_rules.cal layout.gds
# Metal Fill
```

```
calibre -drc -run metalfill metal_fill_rules.cal
# View results
calibredrv -m layout.gds -rve drc.results
```

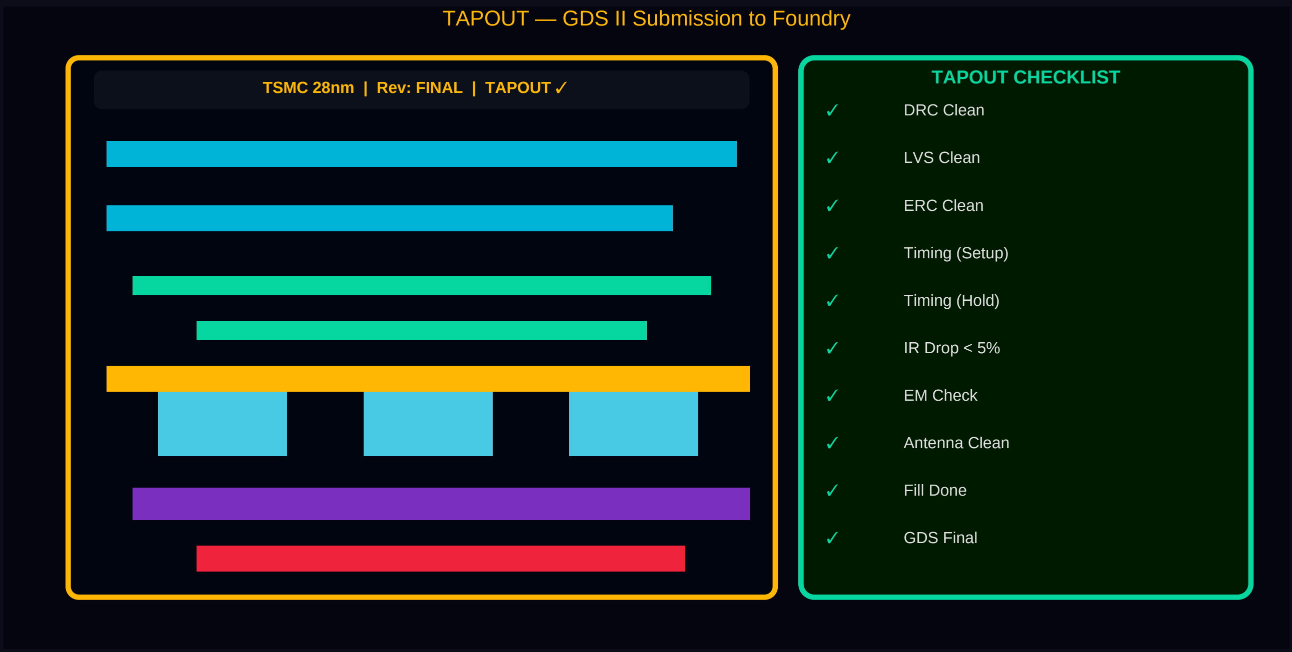
### Real-Time Engineer Notes

- *Never stream out GDS without first running DRC — even 1 DRC error causes foundry rejection.*
- *Common 28nm DRC violations: M1 min-area, via coverage, and well-tap spacing violations.*
- *LVS 'softchk' mode helps debug net mismatches before full LVS — saves run time.*
- *Floating wire DRCs (seen in ORCA TOP): incomplete routing segment — fix in ICC2 with 'fix_route'.*
- *After ECO routing, always re-run full Calibre DRC — ECO routes often introduce new violations.*

### Tapout Relevance

DRC clean and LVS clean are absolute tapout requirements — no exceptions. The foundry performs their own DRC check on received GDS. Any violation results in immediate rejection. LVS mismatch means the fabricated chip will not function. These are the final gates before GDS submission.

# STAGE 9 Tapout — GDS Submission to Foundry




## Overview

Tapout (also called 'tapeout') is the final milestone of the ASIC design flow — the act of submitting the final GDS II database to the semiconductor foundry for fabrication. It is called 'tapout' from the era when design data was written to magnetic tape. A tapout package includes the GDS II file (full chip layout), CDL netlist (for foundry LVS), OASIS/GDS layer map, power intent (UPF), timing sign-off reports, and foundry-specific documentation. The foundry then manufactures photomasks and begins wafer fabrication. A successful tapout requires all prior stages to be signed off with zero critical violations.

## Specifications & Parameters

| | |
|---|---|
| **GDS Format** | GDS II (GDSII) / OASIS — foundry specified |
| **CDL Netlist** | Circuit Description Language — for foundry LVS |
| **Layer Map** | Foundry-specific layer/purpose map file |
| **Technology** | TSMC 28nm HPC / SMIC 32nm (project-specific) |
| **Power Intent** | UPF (Unified Power Format) for multi-voltage |
| **Timing Reports** | PrimeTime sign-off: 0 WNS, 0 TNS, all corners |
| **DRC Status** | 0 violations — Calibre nmDRC |
| **LVS Status** | Clean — Calibre nmLVS |
| **Metal Density** | All layers within foundry density window |
| **Foundry Submission** | Secure portal upload + physical media (NDA-protected) |

## Real-Time Tool Commands (ICC2 / Industry)

```
# Final GDS stream-out from ICC2
write_gds -output chip_final.gds -layer_map tsmc28_layer.map
# OASIS alternative
write_oasis -output chip_final.oas
# Final DRC sign-off
calibre -drc -hier -turbo 16 tsmc28_signoff.cal chip_final.gds > final_drc.log
# LVS sign-off
```

```
calibre -lvs -hier tsmc28_lvs.cal chip_final.cdl chip_final.gds > final_lvs.log
# GDS verification
calibredrv -m chip_final.gds -rc -rve final_drc.results
# Compress for submission
gzip -9 chip_final.gds
```

### Real-Time Engineer Notes

- *Create a tapout checklist — verify every item before submitting: DRC, LVS, STA, IR drop, EM, fill, antenna.*
- *Foundry review takes 2–5 business days — any rejection adds weeks; get it right the first time.*
- *GDS file size at 28nm full chip: 5–50 GB — ensure compression and checksum before upload.*
- *Timing sign-off must include all MCMM corners — missing one corner is a tapout blocker.*
- *At SiValley Technologies: tapout review included 48-hour continuous DRC/LVS re-runs before submission.*
- *Post-tapout: receive MPW slot confirmation, NRE invoice, and estimated silicon delivery (8–14 weeks).*

### Tapout Relevance

Tapout IS the culmination of the entire ASIC flow. Every stage from RTL to routing contributes to tapout readiness. The tapout checklist (DRC clean, LVS clean, STA sign-off, IR drop < 5%, EM rules met, density fill complete, antenna clean, power-on reset verified) must be 100% green. One missed item can result in a non-functional chip, costing $500K–$5M in re-spin costs at advanced nodes.

# ASIC Flow — Complete Sign-Off Summary

| Stage | Key Output | Sign-Off Criteria | Tools |
|---|---|---|---|
| 1. RTL | Verified RTL | Lint clean, CDC clean, Sim coverage >95% | VCS, Spyglass |
| 2. Synthesis | Gate Netlist | 0 WNS, Area budget met, DFT inserted | DC / Genus |
| 3. Floorplan | Placed DEF | IR drop <5%, PG complete, No macro overlap | ICC2 / Innovus |
| 4. Placement | Placed DEF | WNS < -100ps (pre-CTS), Congestion <1% | ICC2 place_opt |
| 5. CTS | Clock Tree | Skew <50ps, Latency met, Hold clean | ICC2 clock_opt |
| 6. Routing | Routed GDS | 0 DRC, 0 shorts/opens, Antenna clean | ICC2 route_opt |
| 7. STA | Timing Rpts | 0 WNS, 0 TNS — all corners all modes | PrimeTime |
| 8. DRC/LVS | Clean GDS | 0 DRC violations, LVS clean, Density met | Calibre |
| **9. TAPOUT** | **GDS II** | **All checks green — submitted to foundry** | **Calibre + ICC2** |

This guide covers the complete ASIC Physical Design Flow from RTL to GDSII tapout with real specifications, industry commands, and lessons from 28nm (ORCA TOP) and 32nm (Raven Wrapper) projects at SiValley Technologies. Each stage's diagram, specification table, TCL commands, and real-time notes reflect hands-on experience with Synopsys ICC2, Fusion Compiler, PrimeTime, and Siemens Calibre.

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